Ogbonna Arinze

arinzeogbo@gmail.com | LinkedIn

SUMMARY

Senior Product Engineer with 8+ years of experience building and shipping end-to-end products across the full stack, from high-end interactive frontends in React and TypeScript to scalable backend systems in Node.js and Python. Proven track record of architecting AI-powered features, agentic workflows, and conversational AI interfaces using LangChain, RAG pipelines, and LLM APIs. Thrives in startup environments with a strong product mindset, driving features from concept through production deployment while collaborating directly with founders and designers on product strategy.

CORE COMPETENCIES

- **Full-Stack Development:** React, TypeScript, Next.js, Node.js, and Python for end-to-end product delivery across frontend and backend.
- **AI & Agent Systems:** LangChain, Vercel AI SDK, Mastra, agentic workflows, conversational AI flows, RAG strategies, embeddings, and vector search (Chroma, Pinecone, Weaviate).
- **Systems Architecture:** Scalable backend design, micro-frontends, monorepos (Turborepo, Nx), SSR/SSG, and event-driven architectures.
- **Real-Time & Streaming:** WebSockets, real-time APIs, low-latency communication, and interactive application development.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, Vercel, GitHub Actions, CI/CD pipelines, and edge rendering.
- **Product & UX Engineering:** Design systems, UI/UX collaboration, accessibility (WCAG), Figma-to-code workflows, and product roadmap contribution.
- **Performance Engineering:** Code splitting, CDN strategies, Core Web Vitals, RUM metrics, bundle optimization, and low-latency delivery.
- **Databases & Data Layer:** PostgreSQL, MongoDB, Redis, Prisma/Drizzle ORM, GraphQL, and REST API design.
- **Quality & Testing:** Jest, Cypress, Playwright, TDD/BDD, automated CI-driven QA, and Allure Reports.
- **Leadership & Collaboration:** Cross-functional collaboration with founders and designers, architecture reviews, mentorship, and Agile methodologies.

WORK EXPERIENCE

Roche Genentech — San Francisco, CA (Remote)

Product Engineer (React / TypeScript / Node.js / Python) — *May 2023 - Present*

- Owned end-to-end feature delivery for a clinical trial data platform, driving features from concept through production deployment by collaborating directly with product managers, designers, and backend engineers on product strategy and technical architecture.
- Architected and built the core **React** and **TypeScript** frontend infrastructure, designing a reusable component library with **Radix UI** that improved UI consistency, accessibility, and developer velocity across multiple product teams.
- Developed **Python** backend services and data processing pipelines for AI-powered features, integrating **LLM APIs** and **agentic workflows** to automate clinical data extraction, classification, and enrichment, reducing manual processing time by over 60%.
- Built **conversational AI interfaces** using **LangChain** and **RAG strategies** with vector search, enabling researchers to query complex clinical trial datasets through natural language, improving data discovery and reducing time-to-insight.
- Designed scalable **Node.js** and **TypeScript** backend systems for real-time data synchronization, implementing **WebSocket** connections for live updates across distributed clinical site integrations, balancing performance with long-term reliability.
- Transformed a legacy **Vue.js** web application into a **React Chrome extension**, optimizing clinical site workflows and enhancing adoption by 40% through close collaboration with design and cross-functional stakeholders.
- Implemented **JWT-based authentication** and **role-based access control** in a **Next.js** fullstack application with **React Query**, using HTTP-only cookies for secure token handling and enforcing permission-based UI rendering across frontend and backend.

- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and efficient **CDN asset delivery** on **AWS CloudFront/S3**, reducing load times by 45% and improving Core Web Vitals scores measured via Lighthouse and Chrome DevTools.
- Scaled **state management** in a large **Next.js** application by integrating **Redux** for complex client state and **React Query** for cached API data, streamlining data flow, cutting redundant re-renders by 50%, and accelerating developer onboarding.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **GitHub Actions CI/CD pipeline**, automating test reporting with Allure Reports and configuring Slack notifications for real-time build quality visibility on every push.
- Drove architectural decisions for the platform's evolution from monolith to **micro-frontend architecture**, defining module boundaries, shared state patterns, and deployment strategies that enabled independent team delivery without regressions.

DESCASIO LTD Lagos, Nigeria
Staff Product Engineer *May 2021 - Apr 2022*

- Operated as a **high-leverage engineer in a startup environment**, owning products end-to-end from concept through deployment, navigating ambiguity and moving fast to ship features that directly impacted revenue and user growth.
- Architected the core **full-stack infrastructure** for a data-heavy **React** dashboard backed by **Node.js** services, implementing virtualization and efficient state management to handle large-scale datasets with seamless real-time updates.
- Built **Python** backend services for automated data processing and analytics, designing RESTful APIs and scheduled data pipelines that processed thousands of records daily, enabling real-time business intelligence reporting.
- Designed and implemented **WebSocket-based real-time features** for live dashboard updates, enabling instant data refresh across connected clients without manual page reloads, improving user engagement and data freshness.
- Shaped **product strategy and technical roadmap** in close collaboration with founders and designers, translating customer feedback into actionable engineering tasks and making trade-offs that balanced speed of delivery with product quality.
- Optimized **frontend and API performance** by profiling render bottlenecks with **React Profiler** and analyzing backend response times, reducing bundle size by **30%-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Deployed and managed application infrastructure on **AWS**, configuring **CloudFront distributions, S3 buckets**, and **Docker** containers for production environments, ensuring high availability and low-latency content delivery.
- Established engineering standards for **code quality, testing**, and deployment across the team using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions CI/CD**, reducing production errors by 40% and accelerating release cycles.
- Integrated **GraphQL APIs** for flexible, efficient data fetching across the dashboard, reducing over-fetching by 60% compared to the previous REST implementation and improving frontend rendering performance.
- Mentored junior engineers on **full-stack best practices**, pair programming, and architecture patterns, fostering a culture of high-quality, maintainable code and consistent design across all product surfaces.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*